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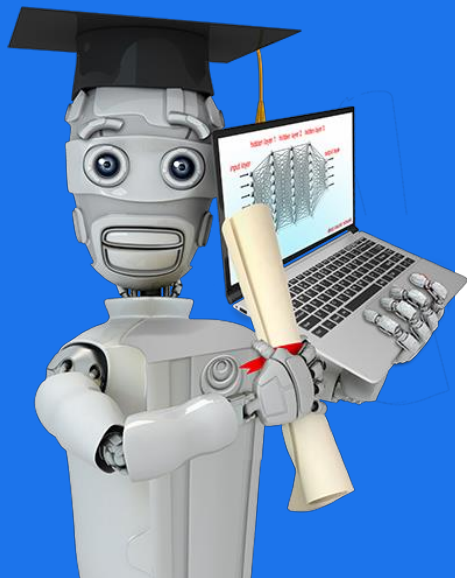
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Classification

Motivations

Classification

Question	Answer " <i>y</i> "	
Is this email <u>spam</u> ?	no	yes
Is the transaction <u>fraudulent</u> ?	no	yes
Is the tumor <u>malignant</u> ?	no	yes

y can only be one of **two** values

"**binary** classification"

class = category

false true

0

1

*useful for
classification*

"negative class"

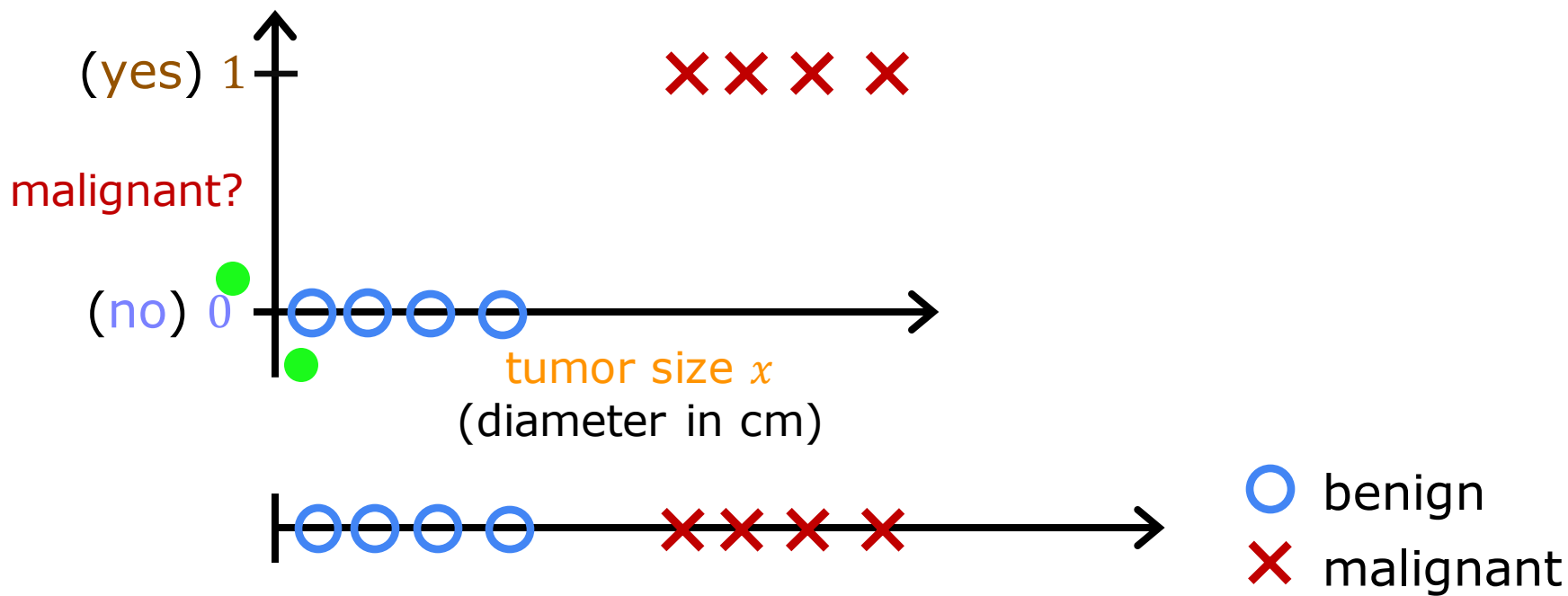
≠ "bad"

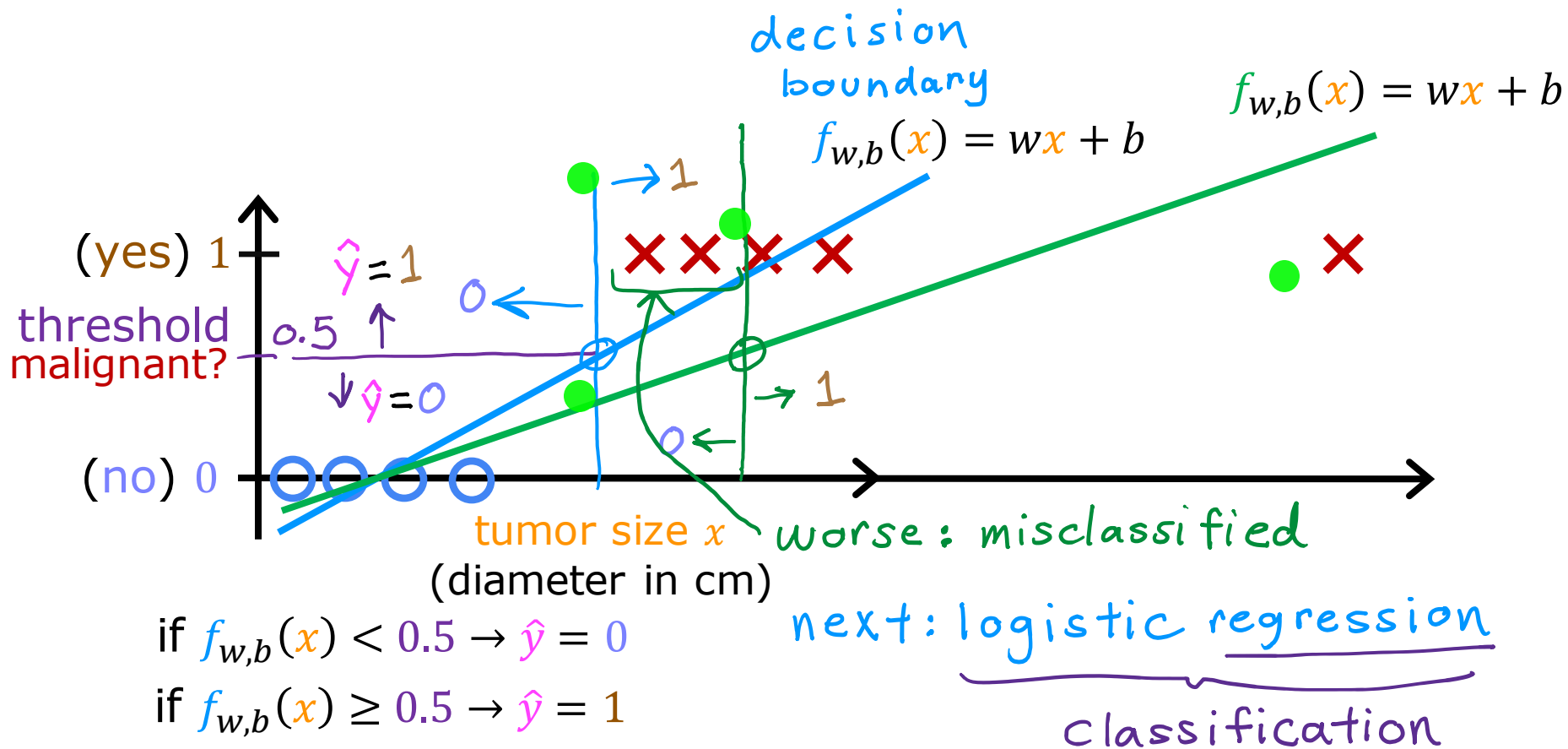
absence

"positive class"

≠ "good"

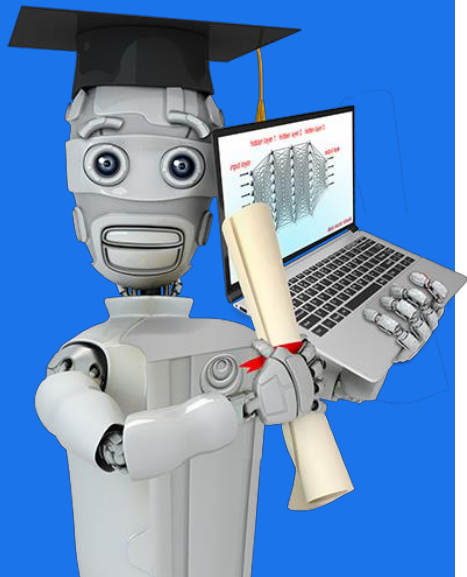
presence





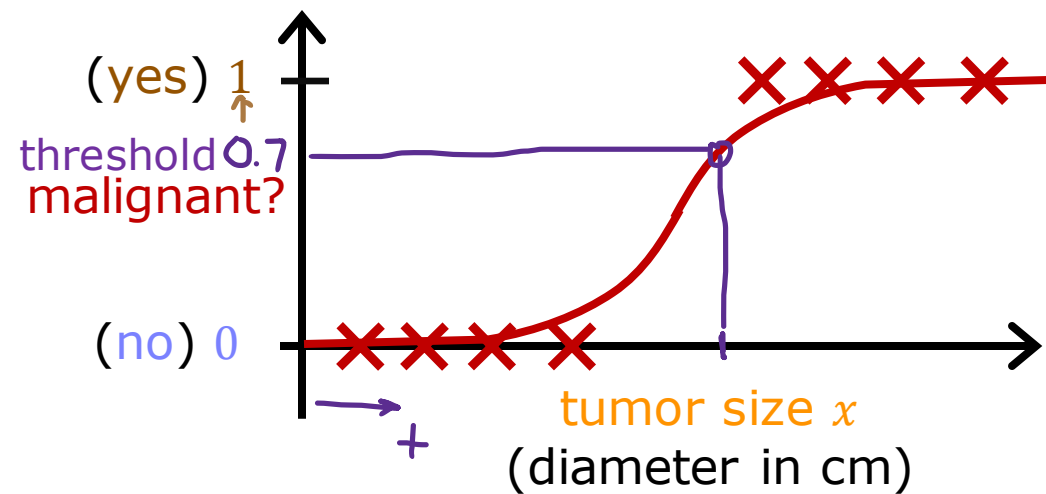
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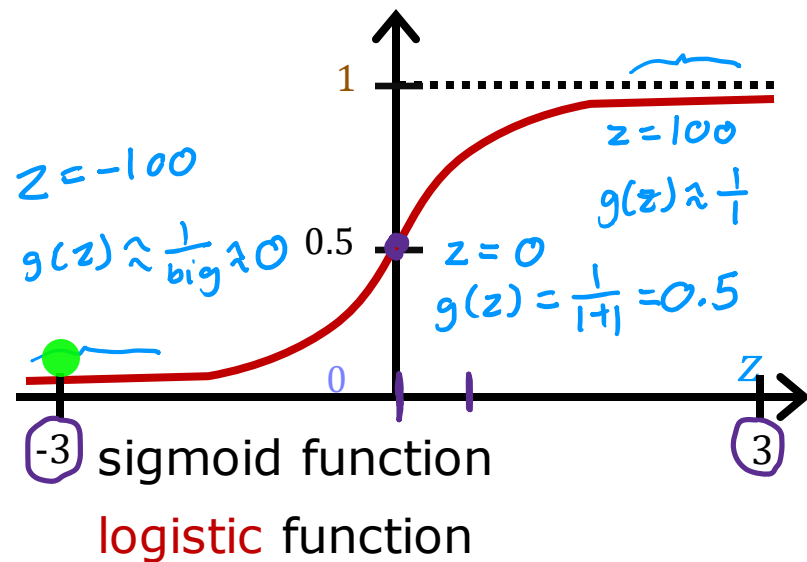


Classification

Logistic Regression



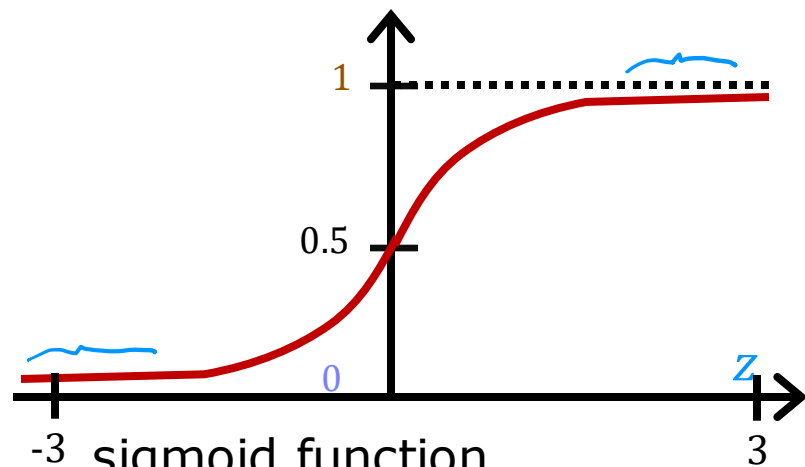
Want outputs between 0 and 1



outputs between 0 and 1

$$g(z) = \frac{1}{1+e^{-z}} \quad 0 < g(z) < 1$$

Want outputs between 0 and 1

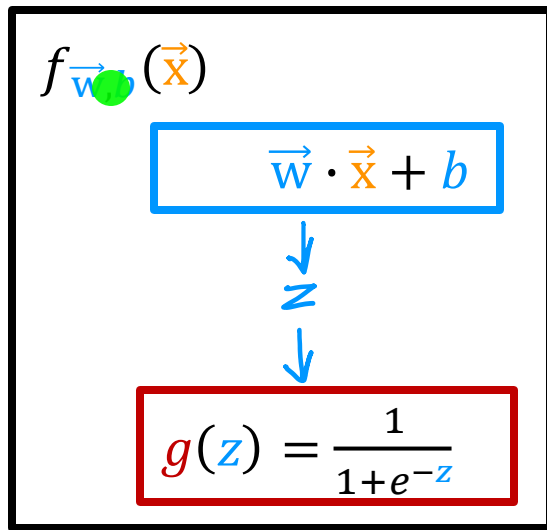


sigmoid function

logistic function

outputs between 0 and 1

$$g(z) = \frac{1}{1+e^{-z}} \quad 0 < g(z) < 1$$



$$f_{\vec{w}, b}(\vec{x}) = g(\underbrace{\vec{w} \cdot \vec{x} + b}_z) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$$

"logistic regression"

$e \approx 2.7$

Interpretation of logistic regression output

$$f_{\vec{w},b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$$

“probability” that class is 1

Example:

x is “tumor size”

y is 0 (not malignant)
or 1 (malignant)

$$f_{\vec{w},b}(\vec{x}) = 0.7$$

70% chance that y is 1

$$f_{\vec{w},b}(\vec{x}) = P(y = 1 | \vec{x}; \vec{w}, b)$$

Probability that y is 1,
given input $\vec{x}$, parameters $\vec{w}, b$

$$P(y = 0) + P(y = 1) = 1$$

🔧 **Table 1: When to Add Polynomial Features (x^2 , x^3 , ...)**

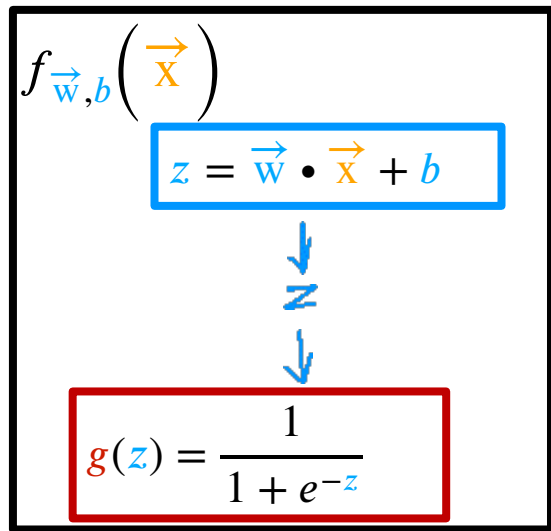
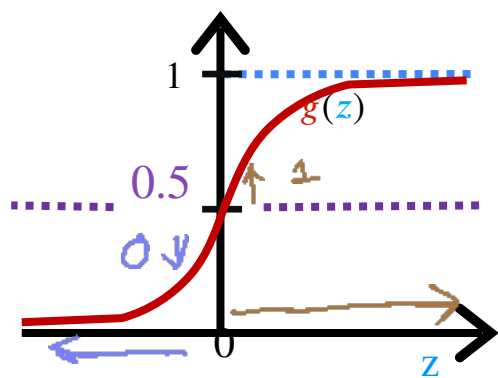
Step	Action	Why / How
1	Start with original features (linear boundary)	Simpler model, easier to interpret
2	Plot data (if low dimension)	Look for non-linear patterns (circles, curves, clusters)
3	Add polynomial features step by step: x_1^2 , x_2^2 , x_1x_2 , then higher powers	Captures curvature, ellipses, interaction effects
4	Use validation / test set to compare performance	Stop when accuracy stops improving
5	If many features, use regularization (L1/ L2)	Prevents overfitting from too many polynomial terms
6	<i>Automated approach:</i> generate degree-2 or degree-3 polynomials and regularize	Common practical method (e.g., <code>PolynomialFeatures</code> in sklearn)

Key takeaway: Let the **data** and **validation error** guide you – don't guess blindly.

Classification

Decision Boundary





$$f_{\vec{w}, b}(\vec{x}) = g(\underbrace{\vec{w} \cdot \vec{x} + \bar{b}}_z) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$$

$$= P(y = 1 | x; \vec{w}, b) \quad 0.7 \quad 0.3$$

0 or 1? threshold

Is $f_{\vec{w}, b}(\vec{x}) \geq 0.5$?

Yes: $\hat{y} = 1$

No: $\hat{y} = 0$

When is

$$f_{\vec{w}, b}(\vec{x}) \geq 0.5 \quad g(z) \geq 0.5$$

$$z \geq 0$$

$$z < 0$$

$$\vec{w} \cdot \vec{x} + b \geq 0$$

$$\vec{w} \cdot \vec{x} + b < 0$$

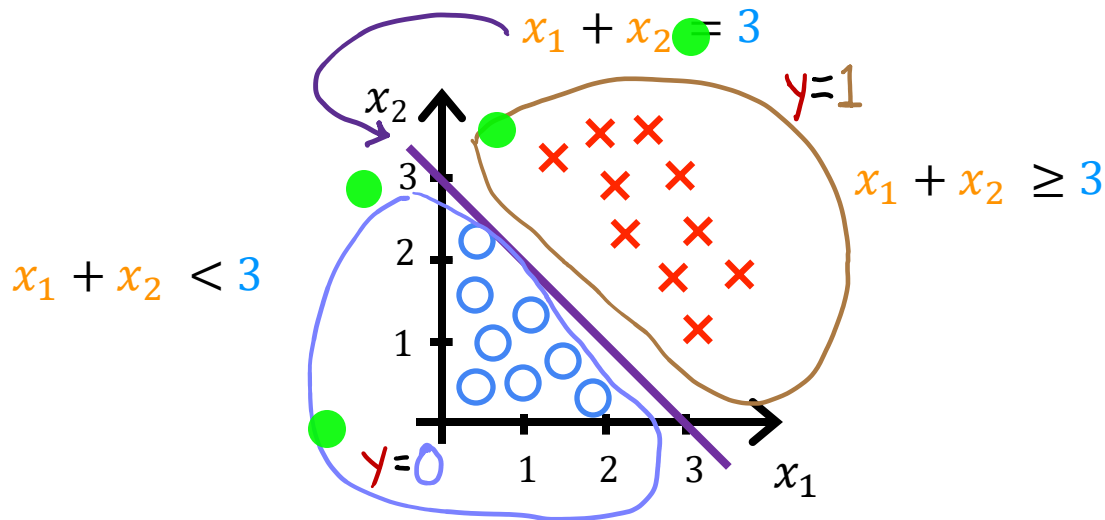
$$\hat{y} = 1$$

$$\hat{y} = 0$$

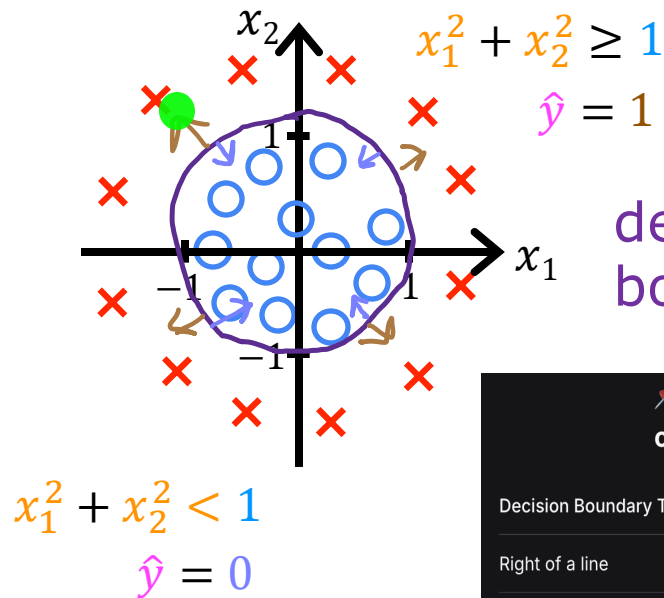
Decision boundary

$$f_{\vec{w},b}(\vec{x}) = g(z) = g(\underbrace{w_1x_1 + w_2x_2 + b}_{z})$$

Decision boundary $z = \vec{w} \cdot \vec{x} + b = 0$
 $z = x_1 + x_2 - 3 = 0$



Non-linear decision boundaries



decision boundary $z = x_1^2 + x_2^2 - 1 = 0$

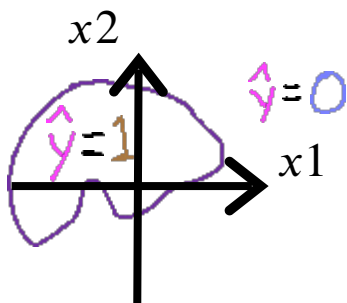
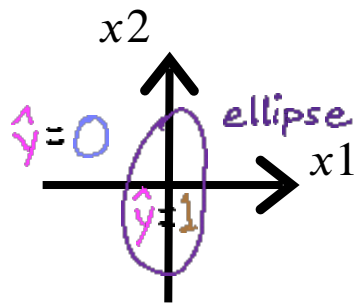
$x_1^2 + x_2^2 = 1$

$$\underbrace{w_1 x_1^2 + w_2 x_2^2 + b}_{z}$$

Table 2: Why Prediction "1" Can Be Outside a Circle or to the Right of a Line

Decision Boundary Type	z Formula	Condition for $z \geq 0 \rightarrow$ Predict $y = 1$	Visual Region
Right of a line	$z = x_1 + x_2 - 3$	$x_1 + x_2 \geq 3$	Right side of the line
Outside a circle	$z = x_1^2 + x_2^2 - 1$	$x_1^2 + x_2^2 \geq 1$	Outside the circle
Inside a circle	$z = 1 - x_1^2 - x_2^2$	$1 - x_1^2 - x_2^2 \geq 0 \rightarrow x_1^2 + x_2^2 \leq 1$	Inside the circle
Top half of a plane	$z = x_2$	$x_2 \geq 0$	Above the x-axis
Irregular shape (blob)	Complex polynomial (e.g., $z = P(x_1, x_2)$)	$P(x_1, x_2) \geq 0$	Outside the blob (if blob encloses negative region)

Non-linear decision boundaries



$$f_{\vec{w},b}(\vec{x}) = g(z) = g(w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_1x_2 + w_5x_2^2 + w_6x_1^3 + \dots + b)$$

📌 Table 3: Quick Reference – Adjusting z to Flip Prediction Region (Bonus)

You want predict 1 here...	Write z as ...	So that $z \geq 0$ there
Right of a line	$x_1 + x_2 - 3$	Right side
Left of a line	$-(x_1 + x_2 - 3)$	Left side
Outside a circle	$x_1^2 + x_2^2 - 1$	Outside
Inside a circle	$1 - x_1^2 - x_2^2$	Inside
Above a line	x_2	Above x-axis
Below a line	$-x_2$	Below x-axis

✓ Summary Table: Log Loss Fundamentals

Concept / Question	Correct / Clarification
$y=1, p=0.1 \rightarrow$ large cost, forces $p \uparrow$	✓ Correct
$y=0, p=0.9 \rightarrow$ large cost, forces $p \downarrow$	✓ Correct
How can p both $\uparrow$ and $\downarrow$ simultaneously?	Parameters are shared; gradient descent combines signals from all examples, and different feature signs produce opposite p movements for different examples.
$\sigma(z) = 1 / (1 + e^{-z})$	✓ Correct (sigmoid function)
$\log(100) = 2 \rightarrow 10^2 = 100$	✓ Correct
$\log_5(9)$ = power of 5 to get 9	≈ 1.365
Why use log in loss?	Turns products into sums; penalizes confident wrong predictions heavily; derived from maximum likelihood.
Log loss (one example) when $y=1$	$-\log(p)$
Log loss (one example) when $y=0$	$-\log(1-p)$

Cost Function

Cost Function for Logistic Regression



Training set

	tumor size (cm) x_1	...	patient's age x_n	malignant? y	$i = 1, \dots, m \leftarrow$ training examples $j = 1, \dots, n \leftarrow$ features
$i=1$	10		52	1	target y is 0 or 1
$\vdots$	2		73	0	
$\vdots$	5		55	0	
	12		49	1	
$i=m$	...		...	...	

$$f_{\vec{w}, b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$$

How to choose $\vec{w} = [w_1 \ w_2 \ \cdots \ w_n]$ and b ?

Squared error cost

cost

$$J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m \left[\frac{1}{2} (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2 \right]$$

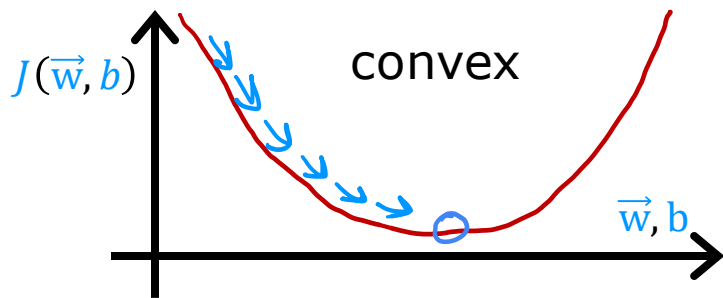
average of training set

loss

$$L(f_{\vec{w}, b}(\vec{x}^{(i)}), y^{(i)})$$

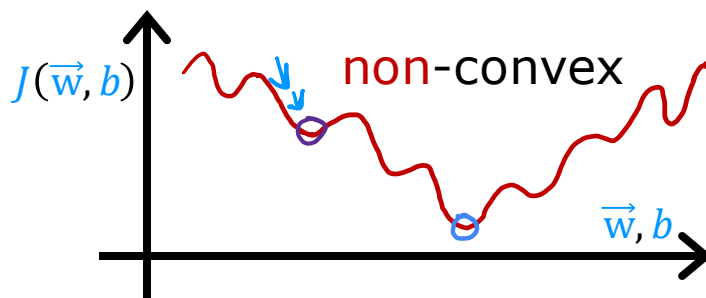
linear regression

$$f_{\vec{w}, b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$$



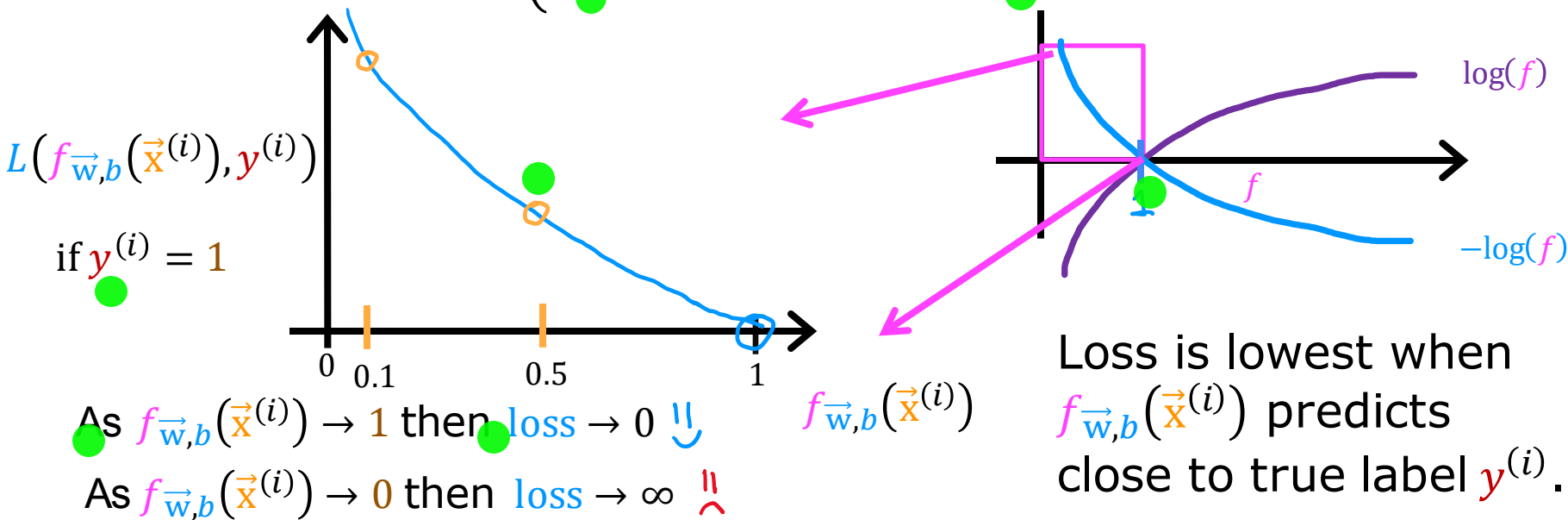
logistic regression

$$f_{\vec{w}, b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$$



Logistic loss function

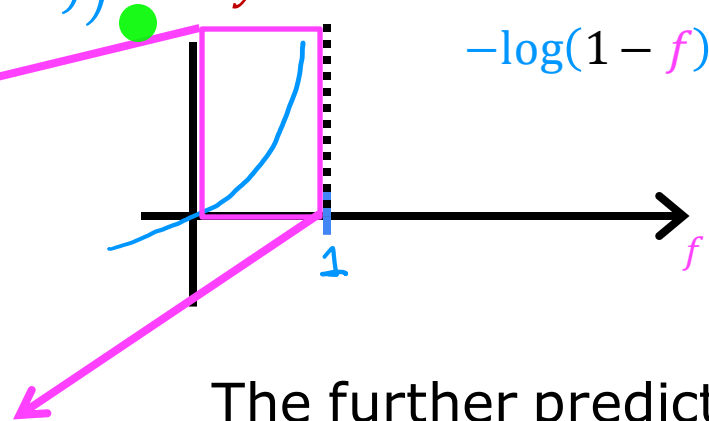
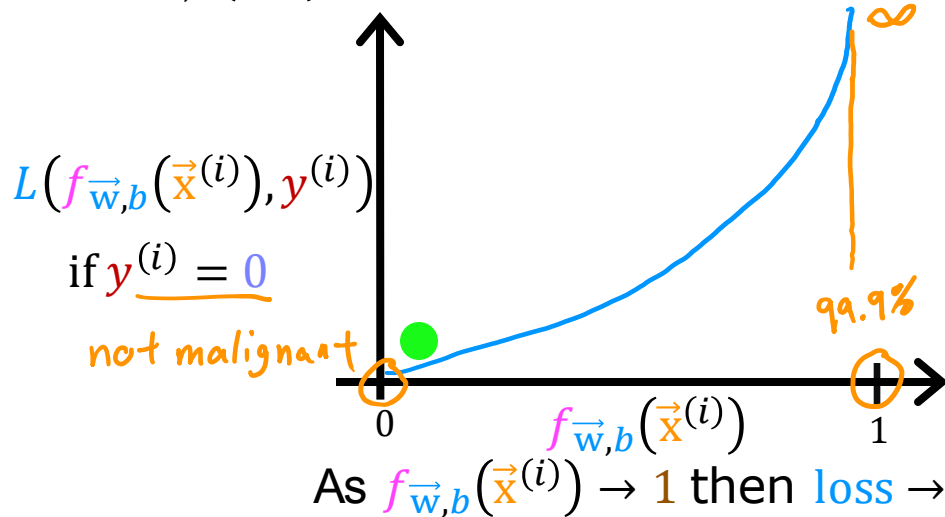
$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = \begin{cases} -\log(f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 1 \\ -\log(1 - f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 0 \end{cases}$$



Logistic loss function

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = \begin{cases} -\log(f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 1 \\ -\log(1 - f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 0 \end{cases}$$

As $f_{\vec{w},b}(\vec{x}^{(i)}) \rightarrow 0$ then $\text{loss} \rightarrow 0$ $\Downarrow$



Cost

$$\text{cost } J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m L(\underbrace{f_{\vec{w}, b}(\vec{x}^{(i)})}_{\text{loss}}, y^{(i)})$$

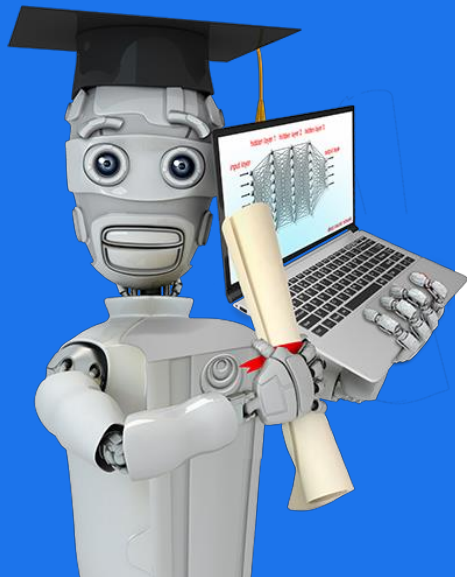
$$= \begin{cases} -\log(f_{\vec{w}, b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 1 \\ -\log(1 - f_{\vec{w}, b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 0 \end{cases}$$

if $y^{(i)} = 1$ *convex* $\rightarrow$ can reach a global minimum
if $y^{(i)} = 0$ *convex* $\rightarrow$ can reach a global minimum

find w, b that minimize cost J

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Cost Function

Simplified Cost Function for Logistic Regression

Simplified loss function

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = \begin{cases} -\log(f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 1 \\ -\log(1 - f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 0 \end{cases}$$

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = - \underbrace{y^{(i)}}_1 \log(f_{\vec{w},b}(\vec{x}^{(i)})) - \underbrace{(1 - y^{(i)})}_{(1-1)=0} \log(1 - f_{\vec{w},b}(\vec{x}^{(i)}))$$

if $y^{(i)} = 1$:

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = -1 \log(f(\hat{x}))$$

Simplified loss function

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = \begin{cases} -\log(f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 1 \\ -\log(1 - f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 0 \end{cases}$$

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = - \underbrace{y^{(i)}}_0 \log(f_{\vec{w},b}(\vec{x}^{(i)})) - (1 - \underbrace{y^{(i)}}_{(1-0)}) \log(1 - f_{\vec{w},b}(\vec{x}^{(i)}))$$

if $y^{(i)} = 1$:

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = -1 \log(f(\vec{x}))$$

if $y^{(i)} = 0$:

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) =$$

$$- (1-0) \log(1 - f(\vec{x}))$$

Simplified cost function

loss

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = - \underbrace{y^{(i)} \log(f_{\vec{w},b}(\vec{x}^{(i)})) + (1 - y^{(i)}) \log(1 - f_{\vec{w},b}(\vec{x}^{(i)}))}_{\text{convex (single global minimum)}}$$

cost

$$J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m [L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)})]$$

$$= \frac{1}{m} \sum_{i=1}^m [y^{(i)} \log(f_{\vec{w},b}(\vec{x}^{(i)})) + (1 - y^{(i)}) \log(1 - f_{\vec{w},b}(\vec{x}^{(i)}))]$$

maximum likelihood
(don't worry about it!)

5. Summary

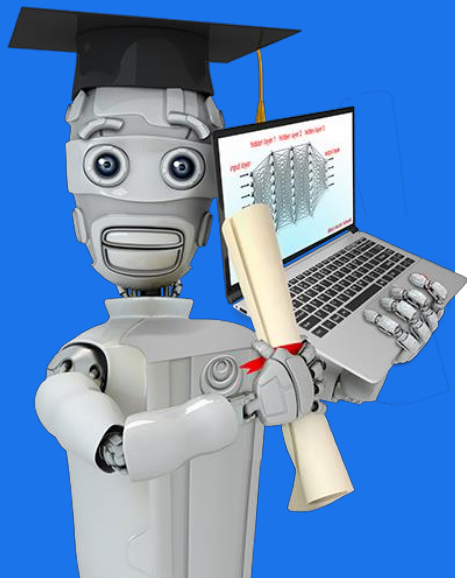
Step	Operation
1	Model probability: $p^{(i)} = \sigma(z^{(i)})$
2	Likelihood for one example: $(p^{(i)})^{y^{(i)}} (1 - p^{(i)})^{1-y^{(i)}}$
3	Likelihood for all: product over i
4	Log-likelihood: sum over i of $y^{(i)} \log p^{(i)} + (1 - y^{(i)}) \log(1 - p^{(i)})$
5	Cost = negative average log-likelihood $\rightarrow J$

So minimizing J is exactly maximizing the likelihood of the training data under the logistic model.

Logistic function ---[produces]--- Probability p log ---[loss]--- Loss per example ---[average over m]--- Cost function

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Gradient Descent

Gradient Descent Implementation

Training logistic regression

Find $\vec{w}, b$

Given new $\vec{x}$, output $f_{\vec{w}, b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$

$$P(y = 1 | \vec{x}; \vec{w}, b)$$

$$p = P(y=1|x;w,b) = \sigma(z) = 1/(1+e^{-z}) \quad \text{— where } z = w \cdot x + b$$

So $p = f(x)$ — the same as $f_{w,b}(x)$ in the lectures.

$$p = f(x) = \sigma(z)$$

Summary Table

Concept	Logistic Regression
Model output during training	Probability $p \in (0, 1)$
Used in gradient	$p - y$ (could be fractional)
Final class prediction	1 if $p \geq 0.5$, else 0

Gradient descent

cost

$$J(\vec{w}, b) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(f_{\vec{w}, b}(\vec{x}^{(i)})) + (1 - y^{(i)}) \log(1 - f_{\vec{w}, b}(\vec{x}^{(i)})) \right]$$

repeat {

$j = 1 \dots n$

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b)$$

} simultaneous updates

$$\frac{\partial}{\partial w_j} J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)}$$
$$\frac{\partial}{\partial b} J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})$$

Gradient descent for logistic regression

repeat {

looks like linear regression!

$$w_j = w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^m (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)} \right]$$

$$b = b - \alpha \left[\frac{1}{m} \sum_{i=1}^m (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}) \right]$$

} simultaneous updates

Same concepts:

- Monitor gradient descent (learning curve)
- Vectorized implementation
- Feature scaling

Linear regression $f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$

Logistic regression $f_{\vec{w},b}(\vec{x}) = \frac{1}{1 + e^{(-\vec{w} \cdot \vec{x} + b)}}$

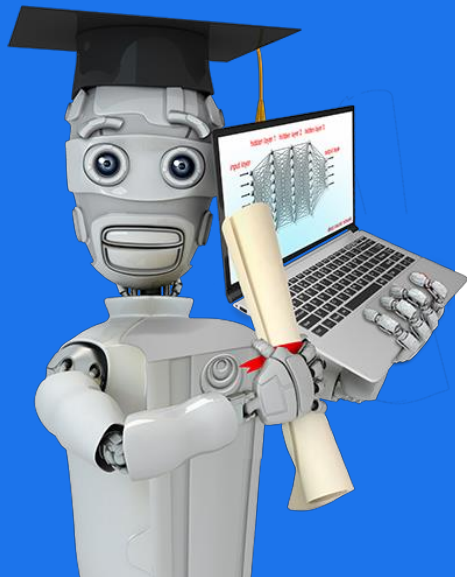
Summary Table

Aspect	Linear Regression	Logistic Regression
Cost function	$\frac{1}{2m} (f - y)^2$	$-[y \log f + (1 - y) \log(1 - f)]$
Derivative of cost w.r.t model output	$f - y$	$f - y$ (after simplification)
Derivative of output w.r.t w_j	x_j	x_j
Final gradient	$\frac{1}{m} \sum (f - y) x_j$	$\frac{1}{m} \sum (f - y) x_j$

Thus, the gradient formula is identical, but the prediction $f(x)$ is what changes between the two algorithms.

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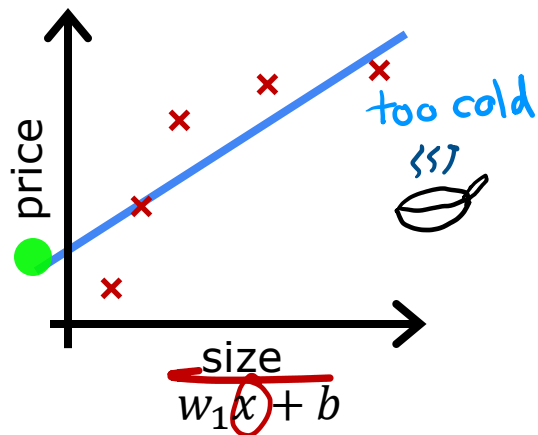
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Regularization to Reduce Overfitting

The Problem of Overfitting

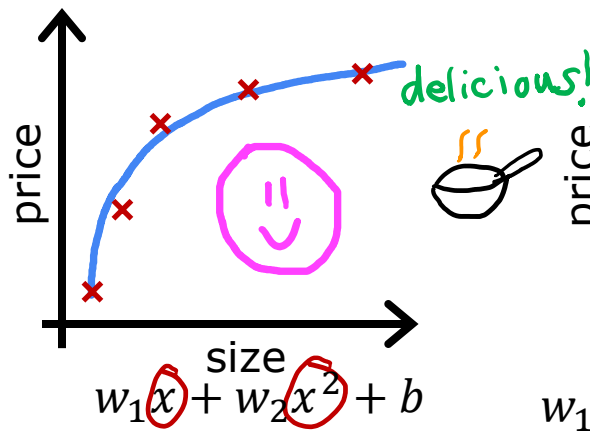
Regression example



underfit

- Does not fit the training set well

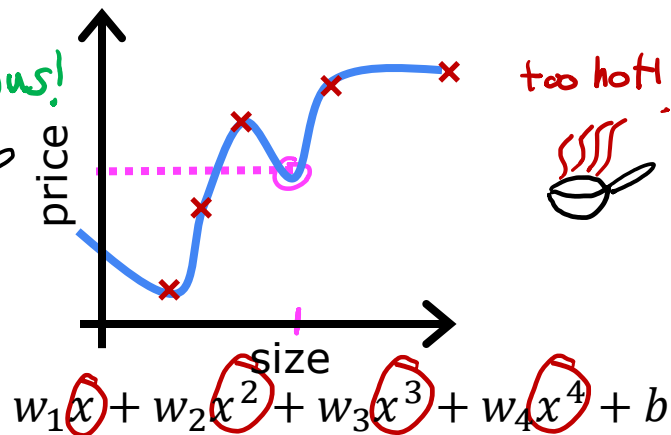
high bias



just right

- Fits training set pretty well

generalization

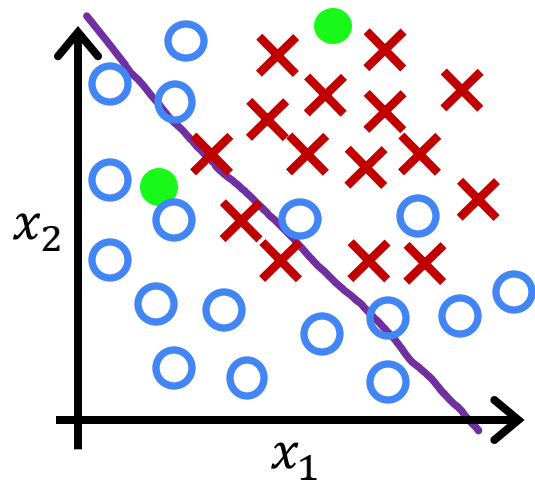


overfit

- Fits the training set extremely well

high variance

Classification

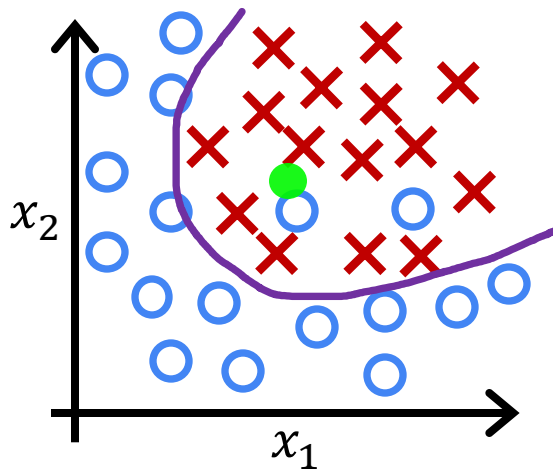


$$z = w_1 x_1 + w_2 x_2 + b$$

$$f_{\vec{w}, b}(\vec{x}) = g(z)$$

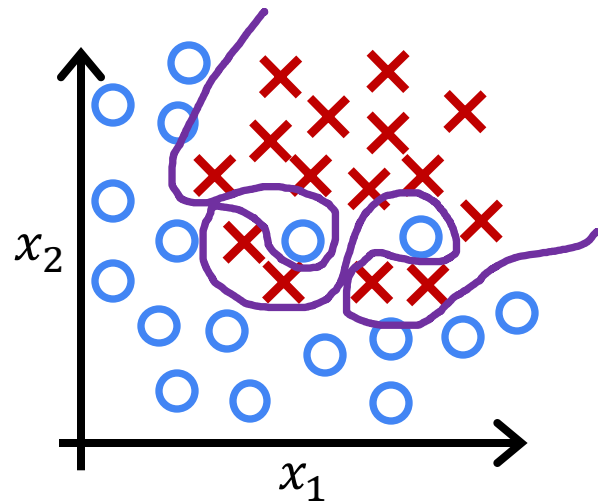
g is the sigmoid function

underfit high bias



$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 + w_4 x_2^2 + w_5 x_1 x_2 + b$$

just right

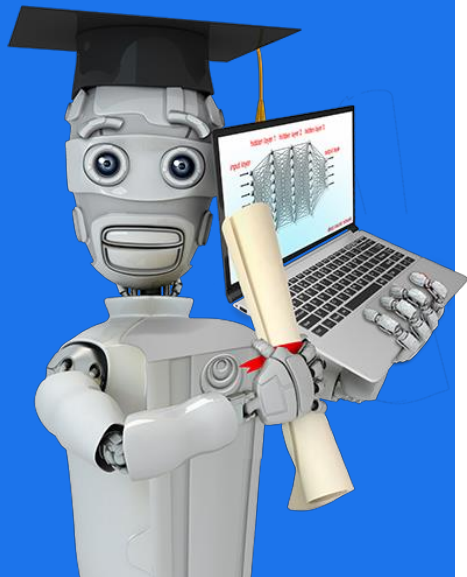


$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 x_2 + w_4 x_1^2 x_2^2 + w_5 x_1^2 x_2^3 + w_6 x_1^3 x_2 + \dots + b$$

overfit

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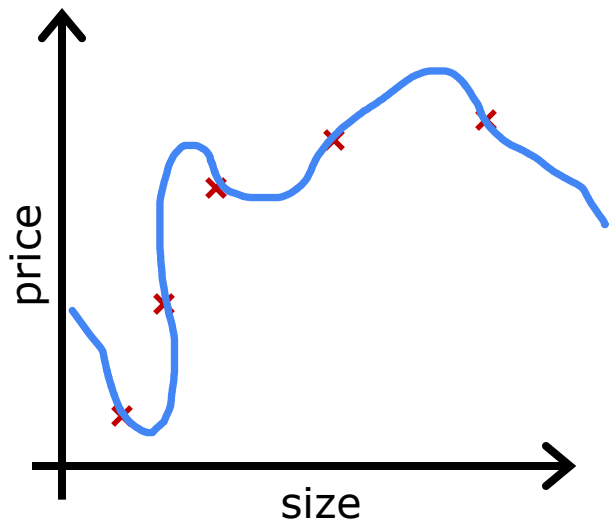
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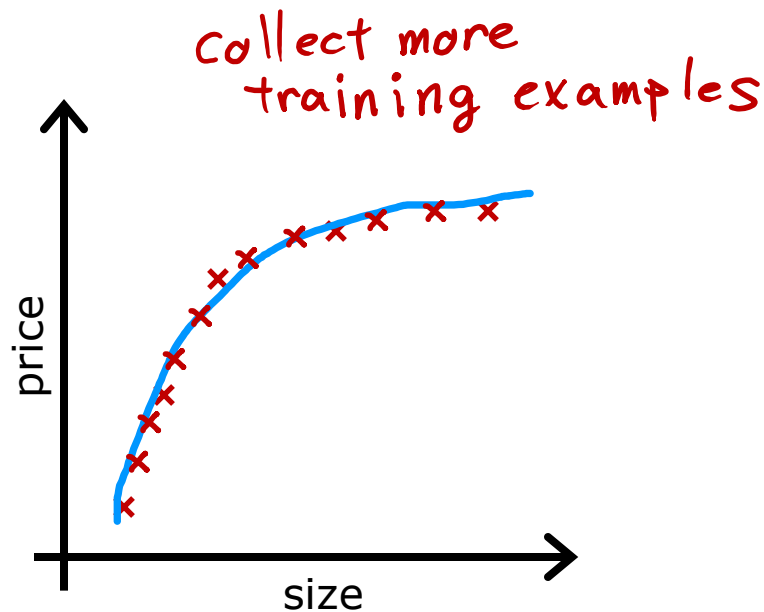
Regularization to Reduce Overfitting

Addressing Overfitting

Collect more training examples



overfit



collect more
training examples

Select features to include/exclude

size	bedrooms	floors	age	avg income	...	distance to coffee shop	price
x_1	x_2	x_3	x_4	x_5		x_{100}	y
●							

all features



insufficient data



overfit

selected features

size
bedrooms
age
just right
feature selection

course 2

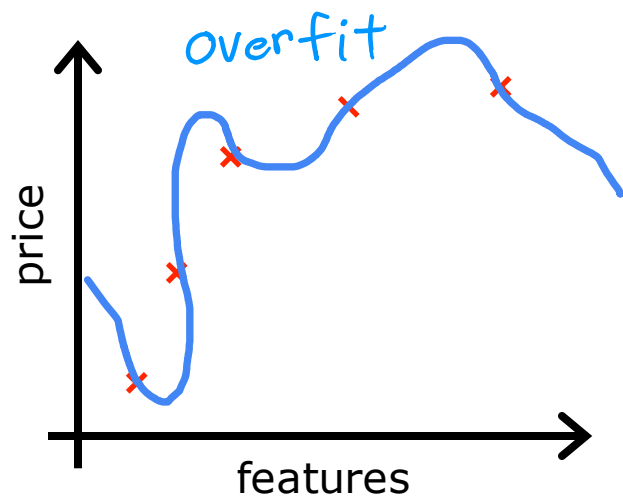
disadvantage



useful features
could be lost

Regularization

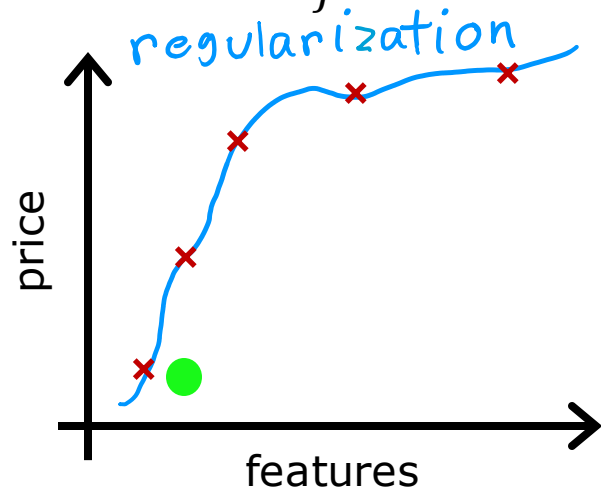
Reduce the size of parameters w_j



$$f(x) = 28x - 385x^2 + 39x^3 - 174x^4 + 100$$

large values for w_j

eliminate feature



$$f(x) = 13x - 0.23x^2 + 0.000014x^3 - 0.0001x^4 + 10$$

small values for w_j

Addressing overfitting

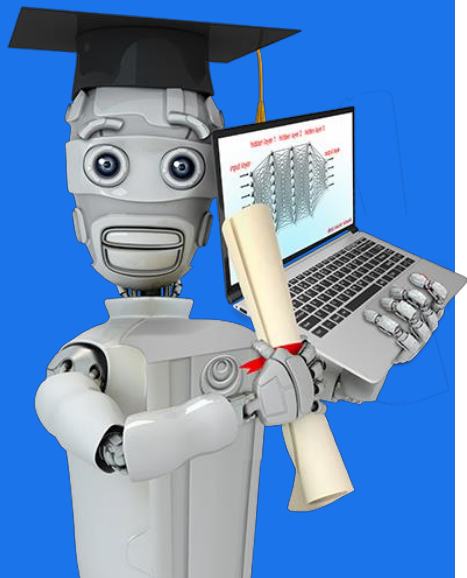
Options

1. Collect more data
2. Select features
 - Feature selection *in course 2*
3. Reduce size of parameters
 - “Regularization” *next videos!*



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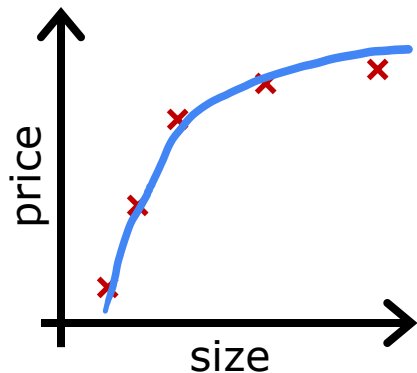
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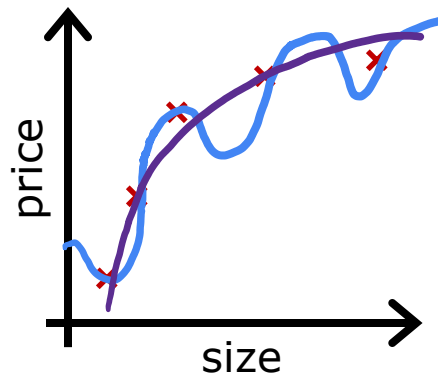
Regularization to Reduce Overfitting

Cost Function with Regularization

Intuition



$$w_1x + w_2x^2 + b$$



$$w_1x + w_2x^2 + \underbrace{w_3x^3}_{\approx 0} + \underbrace{w_4x^4}_{\approx 0} + b$$

make w_3, w_4 really small (≈ 0)

$$\min_{\vec{w}, b} \frac{1}{2m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2 + \cancel{1000 \underbrace{w_3^2}_{0.001}} + \cancel{1000 \underbrace{w_4^2}_{0.002}}$$

Regularization

small values $w_1, w_2, \dots, w_n, b$

simpler model

less likely to overfit

$$w_3 \approx 0$$

$$w_4 \approx 0$$

size x_1	bedrooms x_2	floors x_3	age x_4	avg income x_5	...	distance to coffee shop x_{100}	price y
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$$w_1, w_1, w_2, \dots, w_{100}, b$$

n features

$n = 100$

$$J(\vec{w}, b) = \frac{1}{2m} \left[\sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2 + \underbrace{\frac{\lambda}{2m} \sum_{j=1}^n w_j^2}_{\text{"lambda" regularization parameter}} + \underbrace{\frac{\lambda}{2m} b^2}_{\text{can include or exclude } b} \right]$$

$\lambda > 0$

Regularization

$$\min_{\vec{w}, b} J(\vec{w}, b) = \min_{\vec{w}, b} \left[\underbrace{\frac{1}{2m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2}_{\text{mean squared error}} + \underbrace{\frac{\lambda}{2m} \sum_{j=1}^n w_j^2}_{\text{regularization term}} \right]$$

fit data

Keep w_j small

λ balances both goals

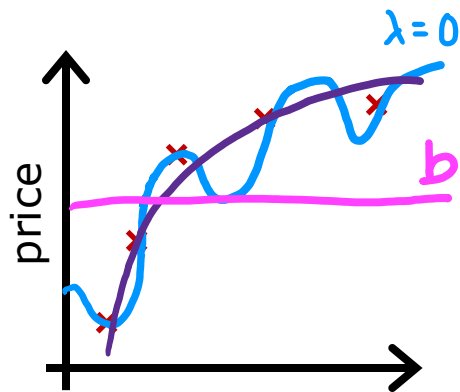
choose $\lambda = 10^{10}$

$$f_{\vec{w}, b}(\vec{x}) = \cancel{w_1}x + \cancel{w_2}x^2 + \cancel{w_3}x^3 + \cancel{w_4}x^4 + b$$

≈ 0 ≈ 0 ≈ 0 ≈ 0

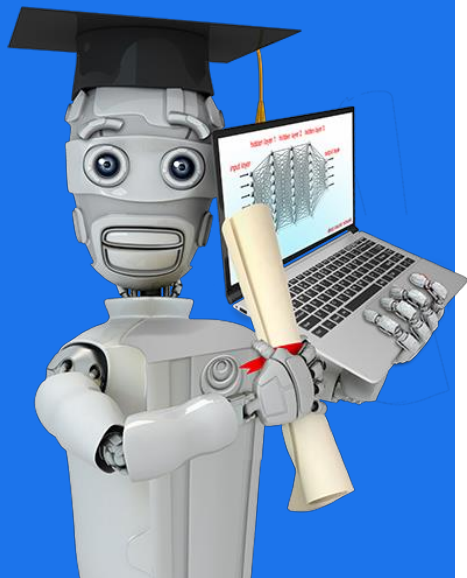
$$f(x) = b$$

choose λ



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Regularization to Reduce Overfitting

Regularized Linear Regression

Regularized linear regression

$$\min_{\vec{w}, b} J(\vec{w}, b) = \min_{\vec{w}, b} \left[\frac{1}{2m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2 \right]$$

Gradient descent

repeat {

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)} + \frac{\lambda}{m} w_j$$

$j = 1, \dots, n$

$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})$$

} simultaneous update

don't have to regularize b

Implementing gradient descent

repeat {

- $$w_j = w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^m \left[(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)} \right] + \frac{\lambda}{m} w_j \right]$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^m (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)})$$

} simultaneous update $j = 1 \dots n$



Summary Table with Full Formulas

Concept	Linear Regression	Logistic Regression
Original cost	$\frac{1}{2m} \sum_{i=1}^m (f(x^{(i)}) - y^{(i)})^2$	$-\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log f(x^{(i)}) + (1 - y^{(i)}) \log(1 - f(x^{(i)}))]$
Regularization term	$\frac{\lambda}{2m} \sum_{j=1}^n w_j^2$	same
Gradient w.r.t w_j	$\frac{1}{m} \sum_{i=1}^m (f(x^{(i)}) - y^{(i)}) x_j^{(i)} + \frac{\lambda}{m} w_j$	same (but f is sigmoid)
Gradient w.r.t b	$\frac{1}{m} \sum_{i=1}^m (f(x^{(i)}) - y^{(i)})$	same
$f(x)$	$w \cdot x + b$	$\sigma(w \cdot x + b) = \frac{1}{1 + e^{-(w \cdot x + b)}}$



Implementing gradient descent

repeat {

$$w_j = w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^m \left[(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)} \right] + \frac{\lambda}{m} w_j \right]$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^m (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)})$$

} simultaneous update $j = 1 \dots n$

$$w_j = \underbrace{w_j - \alpha \frac{\lambda}{m} w_j}_{w_j \left(1 - \alpha \frac{\lambda}{m} \right)} - \underbrace{\alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)}}_{\text{usual update}}$$

shrink w_j

$$\alpha \frac{\lambda}{m} = 0.01 \frac{1}{50} = 0.0002$$

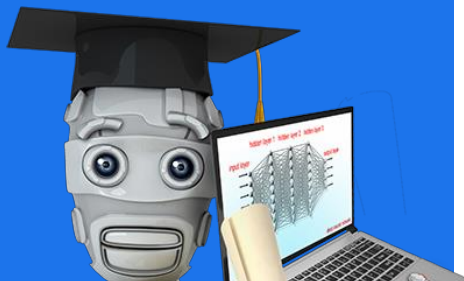
$$w_j (1 - 0.0002) = 0.9998 w_j$$

How we get the derivative term (optional)

$$\begin{aligned}
 \bullet \frac{\partial}{\partial w_j} J(\vec{w}, b) &= \frac{d}{dw_j} \left[\frac{1}{2m} \sum_{i=1}^m \underbrace{(f(\vec{x}^{(i)}) - y^{(i)})^2}_{\vec{w} \cdot \vec{x}^{(i)} + b} + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2 \right] \\
 &= \cancel{\frac{1}{2m} \sum_{i=1}^m} \left[(\vec{w} \cdot \vec{x}^{(i)} + b - y^{(i)}) \cancel{2} x_j^{(i)} \right] + \cancel{\frac{\lambda}{2m}} \cancel{2} w_j \quad \text{No } \sum_{j=1}^n \\
 &= \frac{1}{m} \sum_{i=1}^m \left[\underbrace{(\vec{w} \cdot \vec{x}^{(i)} + b - y^{(i)})}_{f(\vec{x})} x_j^{(i)} \right] + \frac{\lambda}{m} w_j \\
 \bullet &= \frac{1}{m} \sum_{i=1}^m \left[(f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)} \right] + \frac{\lambda}{m} w_j
 \end{aligned}$$

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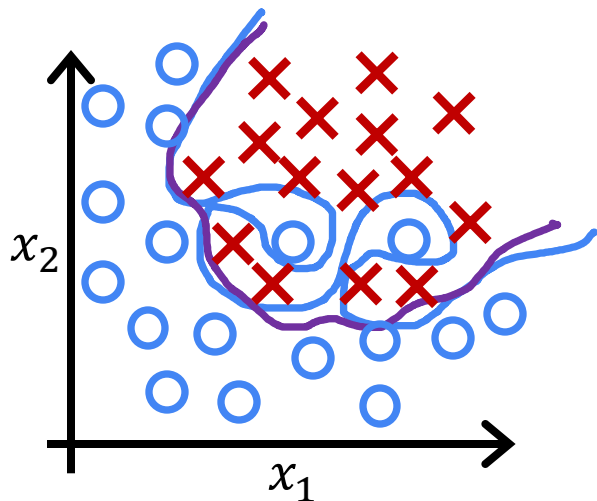
Regularization to Reduce Overfitting

Regularized Logistic Regression

Summary Table

Name	Penalty function	Parameter	Effect on weights
L2 (Ridge)	$\sum w_j^2$	λ (or λ_2)	Shrinks all, none to zero
L1 (Lasso)	$\sum \ w_j\ $	λ (or λ_1)	Some become exactly zero
Elastic Net	$\lambda_1 \sum \ w_j\ + \lambda_2 \sum w_j^2$	λ_1, λ_2	Mix of both

Regularized logistic regression



$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 x_2 + w_4 x_1^2 x_2^2 + w_5 x_1^2 x_2^3 + \dots + b$$

$$f_{\vec{w},b}(\vec{x}) = \frac{1}{1 + e^{-z}}$$

Cost function

$$J(\vec{w}, b) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(f_{\vec{w},b}(\vec{x}^{(i)})) + (1 - y^{(i)}) \log(1 - f_{\vec{w},b}(\vec{x}^{(i)})) \right] + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2$$

$\min_{\vec{w}, b} J(\vec{w}, b) \rightarrow w_j \downarrow$

Regularized logistic regression

$$J(\vec{w}, b) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(f_{\vec{w}, b}(\vec{x}^{(i)})) + (1 - y^{(i)}) \log(1 - f_{\vec{w}, b}(\vec{x}^{(i)})) \right] + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2$$

min $\vec{w}, b$

Gradient descent

repeat {

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$$

j = 1...n

$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b)$$

}

*Looks same as
for linear regression!*

$$= \frac{1}{m} \sum_{i=1}^m \left[(f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)} \right] + \frac{\lambda}{m} w_j$$

logistic regression

*don't have to
regularize b*